Hypothesis

Exposure to the Om sound will decrease stress levels in *Daphnia magna*.

Figure 1. Image taken by researcher of Petri dish with *Daphnia magna*

Purpose

Stress can be a huge problem! This study investigates an ancient Indian remedy Om as a treatment to reduce stress using *Daphnia magna* (water fleas) as a model organism.

Background

- Cardiovascular strain and several health problems are caused by stress
- OM is a low frequency sound that can be generated using a Tibetan singing bowl and is often used to help a person feel calm
- OM is considered to be a sacred and universal vibration
- OM creates a sustained deep, resonant drone sound
- It is always used when chanting the name of Bhagwan (meaning God in Sanskrit)
- Heart rate levels in *Daphnia* was measured when Om treatments were administered for various lengths of time

Procedures

- 16 *Daphnia* were tested
- *Daphnia* was placed on concave lens under microscope
- Heart beat was measured and recorded for 60 seconds
- *Daphnia* was exposed to 20 seconds Om sound
- Heart beat was recorded for 60 seconds duration
- Difference in rates was calculated and recorded
- Procedures were repeated at 2 day intervals for 15 sec and 10 seconds respectively

Does the Low Frequency OM Sound Affect Stress Levels in *Daphnia magna?*

Results

Trial run 1: Average heart rate before 20 sec of Om was 105 BPM and after Om treatment was 63 BPM (40% decrease).
Trial run 2: Average heart rate before 15 sec of Om was 127 BPM and after Om treatment was 92 BPM (27% decrease).
Trial run 3: Average heart rate before Om was 145 BPM and after Om treatment was 97 BPM (33% decrease).

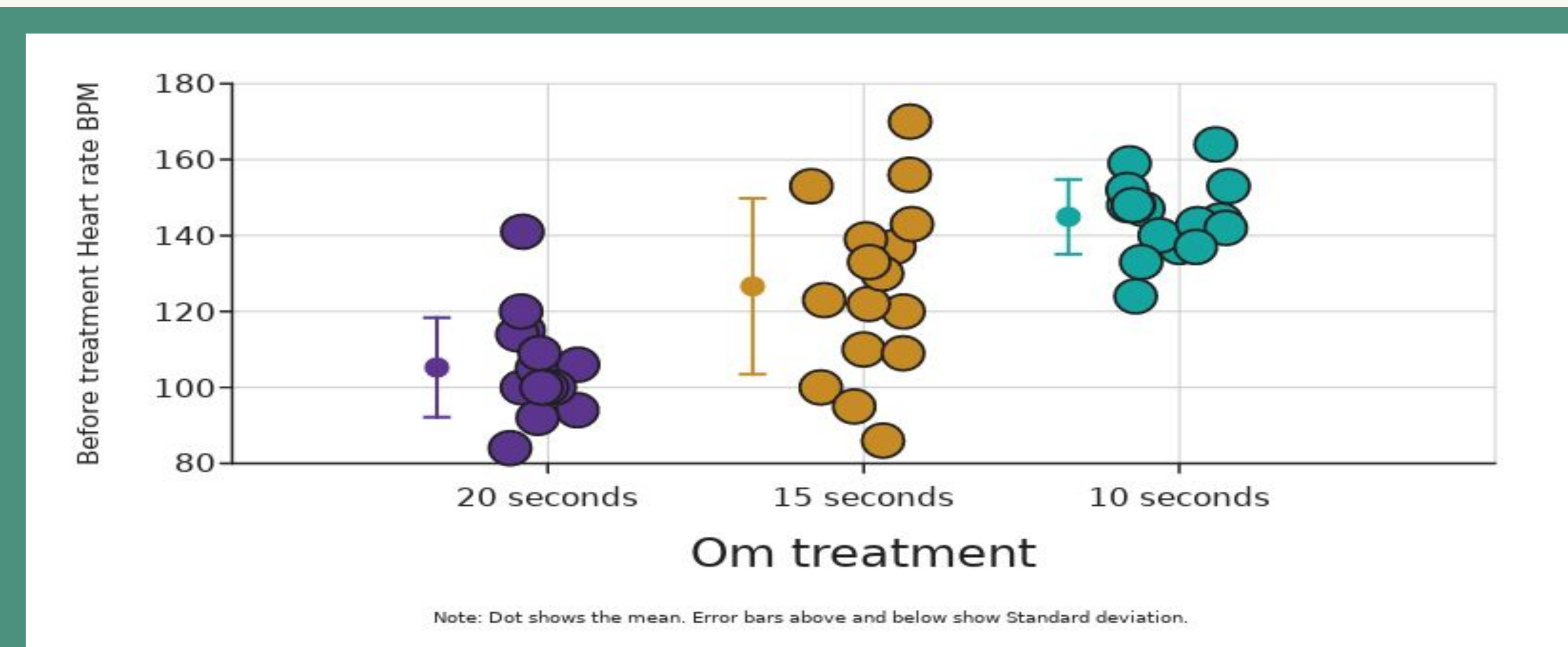


Figure 2. Graph made by researcher using DataClassroom showing the heart rate of *Daphnia magna* before the Om treatment. (scatter plot.)

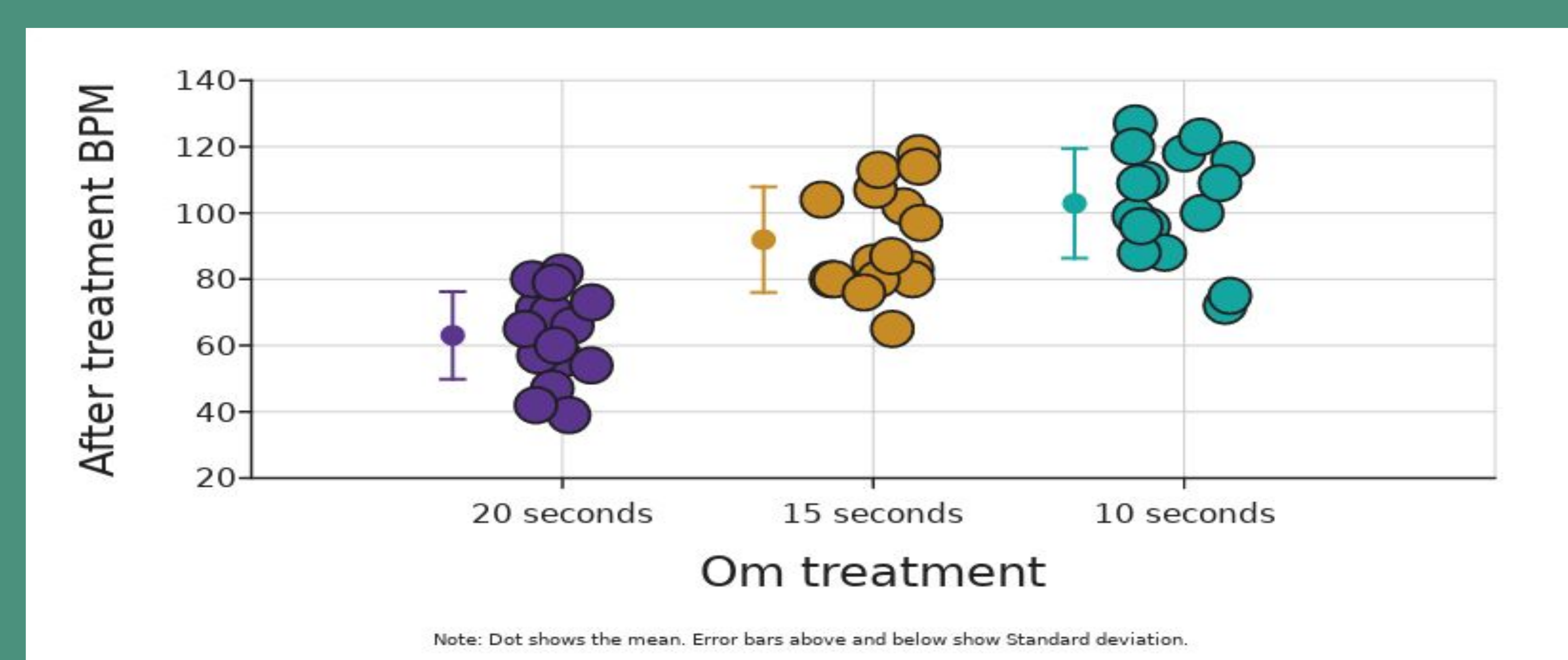


Figure 3. Graph made by researcher using DataClassroom showing the heart rate of *Daphnia magna* after Om treatment. (scatter plot)

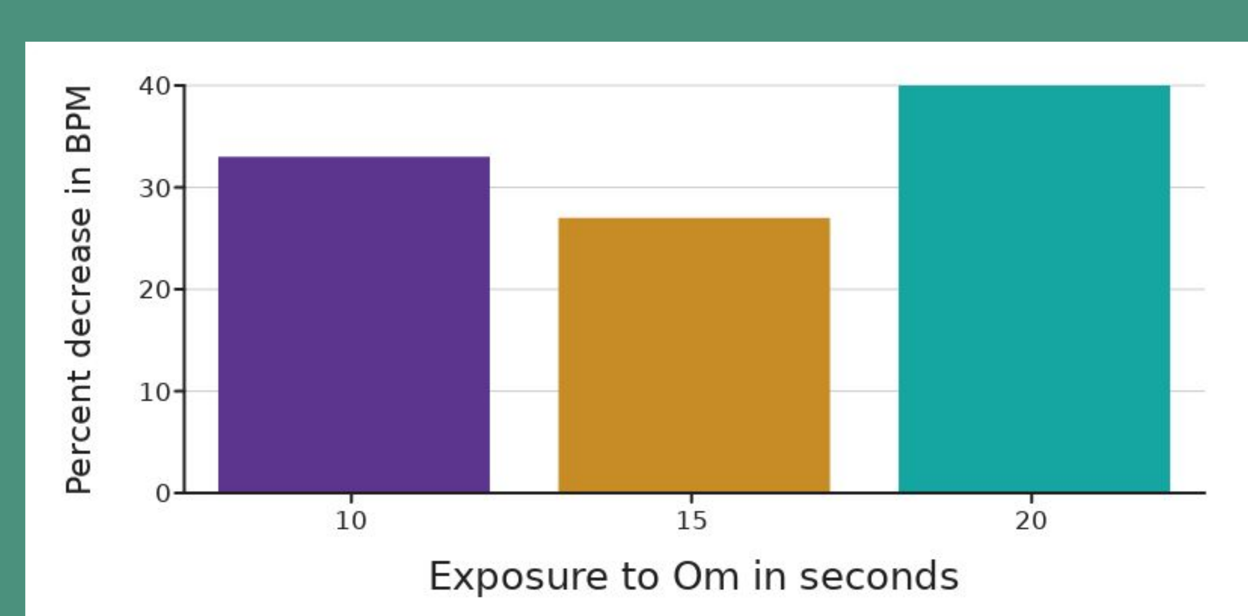
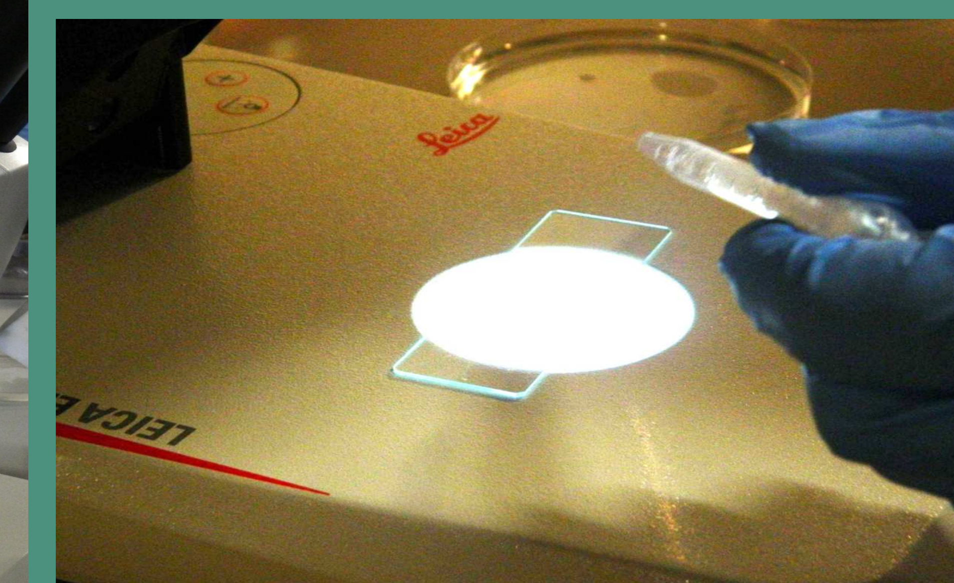


Figure 4. Graph made by researcher using DataClassroom showing the heart rate of *Daphnia magna* showing the percent decrease of heart rate when being exposed to various intervals of the Om treatment

Conclusion

Based on the data, the Om sound affected the *Daphnia magna* by decreasing heart rate/stress level. The 20 second Om treatment had the highest impact on reducing stress on *Daphnia magna* with the average of 63 BPM after treatment indicated a decrease of 40%. This shows that the Om sound had an effect on lowering stress levels in organisms. In the future, humans may use the Om sound to cope with stress.



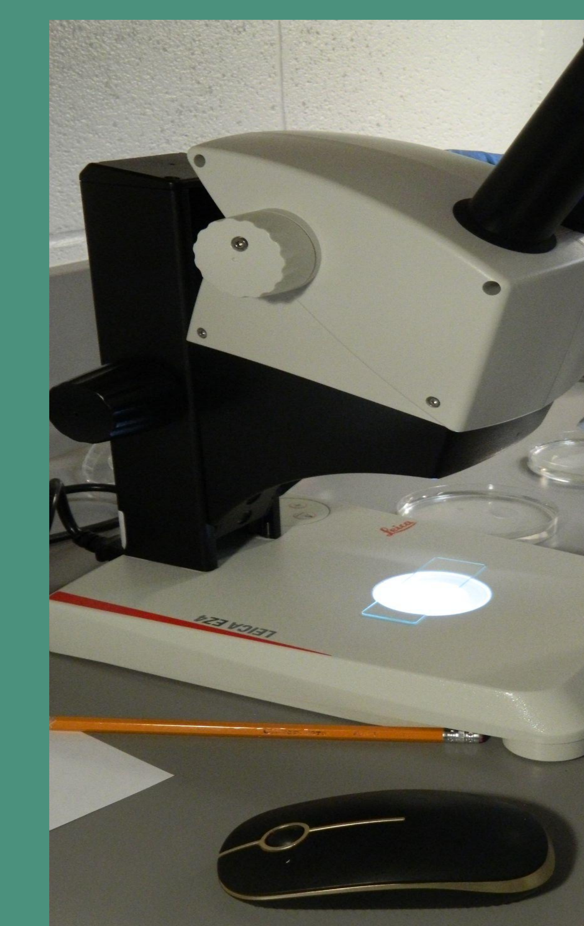


Figure 5. Image taken by researcher of the Pipetting of *Daphnia magna*

Figure 6. Image taken by researcher of a microscope on *Daphnia magna*

Figure 7. Image taken by researcher of *Daphnia magna* under the microscope

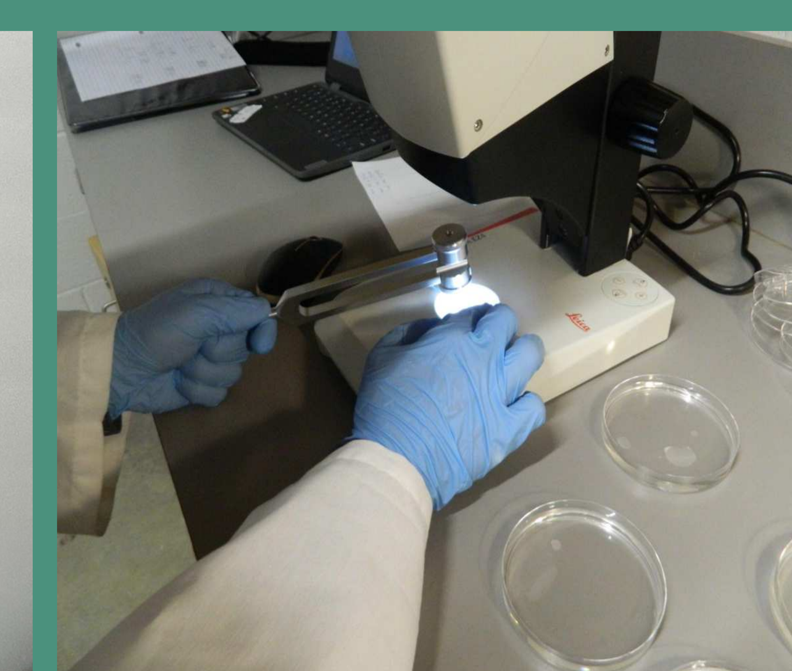


Figure 8. Image taken by researcher of the Om treatment of *Daphnia magna*

References

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